

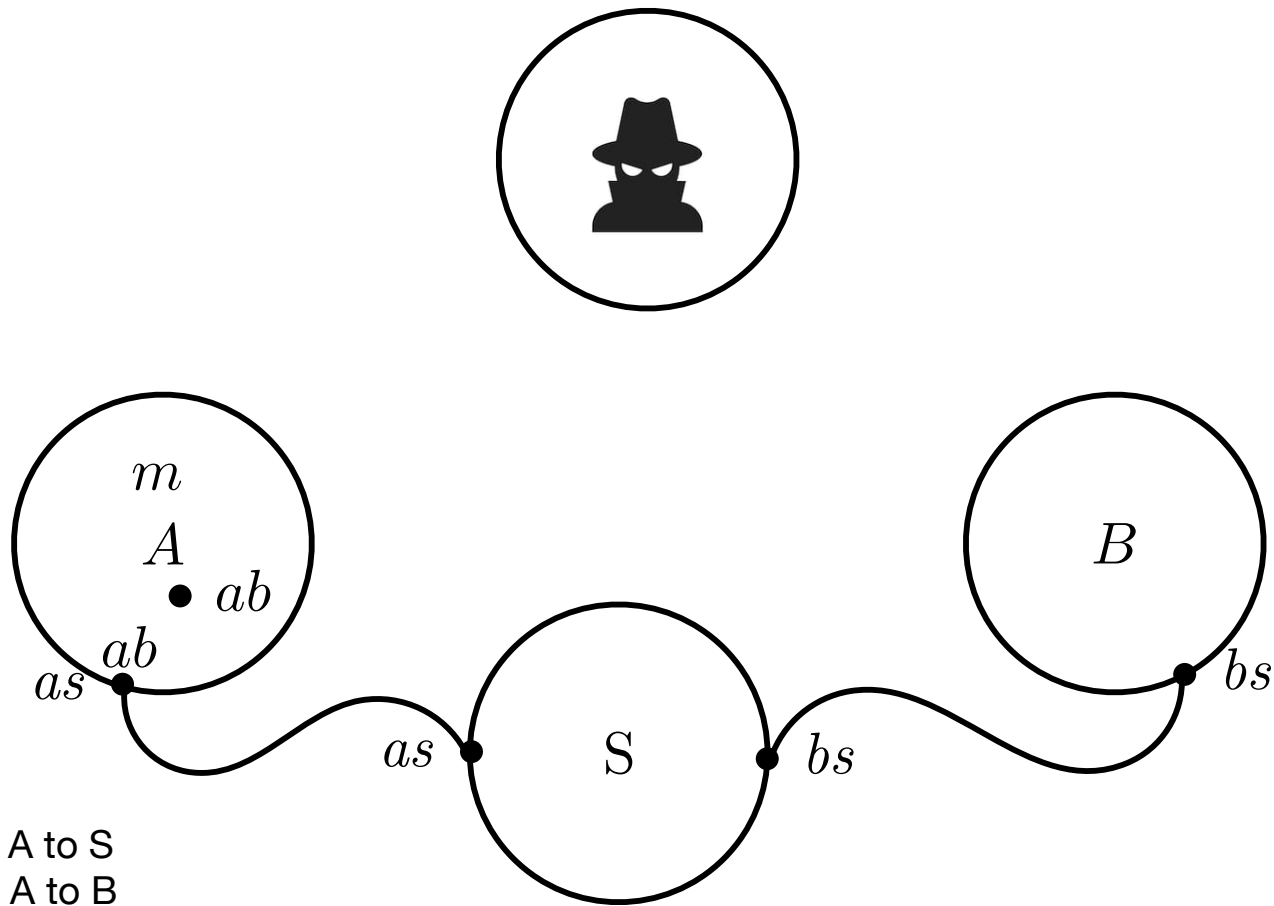
0-CFA for process algebras

Name mobility

- You should have seen CCS
- In CCS communication is statically defined: channels cannot be transmitted / received
- What if channels are treated as ordinary values?
- Can we extend CCS to communicate communication means?
- Yes, this leads us to π -calculus

π -calculus

Name mobility



Legenda:

m: message

as: secure channel from A to S

bs: secure channel from A to B

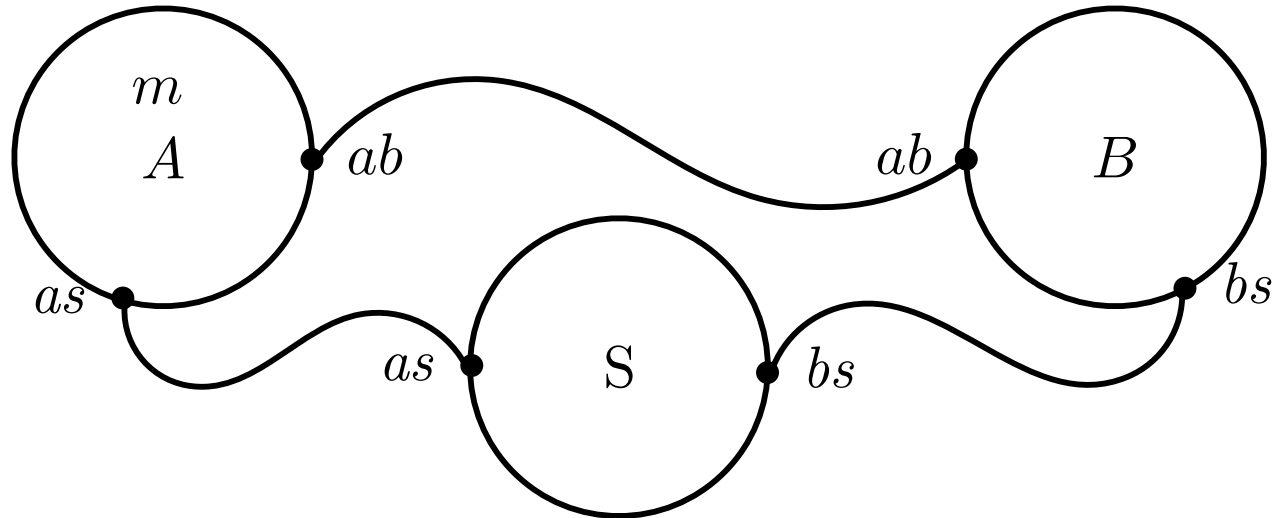
ab: channel from A to B

Name mobility

$(\nu as)(\nu bs)((\nu ab) A | B | S)$



$(\nu as)(\nu bs)((\nu ab)(A | B) | S)$



π -calculus: syntax

Prefixes

$\pi ::= \bar{u}\langle v \rangle$ // send v on u

| $u(x)$ // receive on u and store in v

| τ // internal action

Processes

$P ::= \mathbf{0}$ // inactive process

often
omitted

| $\pi . P$ // action prefix

| $[x = y]P$ // matching (like, if $x = y$ then P)

| $(\nu n)P$ // restriction (like n only known in P)

sometimes
written $P \setminus n$

| $\sum_{i \in I} \pi_i . P_i$ // non deterministic, guarded choice

| $P_1 \mid P_2$ // parallel composition

| $!P$ // replication (like $P \mid P \mid P \mid P \mid \dots$)

nullary sum: $\mathbf{0}$
binary sum: $\pi_1 . P_1 + \pi_2 . P_2$

π -calculus: the idea

- Communication primitives: simple and powerful
- Scoping Rules: explicitly control the access to channels and to data

send z on x

Communication (on channel x)

$$\bar{x}\langle z \rangle . P \mid x(y) . Q \rightarrow P \mid Q[z/y]$$

receive y from x

y has been substituted for z in the second process

Extrusion of z

$$(\nu z) (\bar{x}\langle z \rangle . P) \mid x(y) . Q \rightarrow (\nu z) (P \mid Q[z/y])$$

- y has been substituted for z in the second process
- the scope of z has been extended to include both processes

Free and bound names

A central component of the π -calculus is that of names
The only binders are input prefix and name restriction

$\pi ::=$

$| \bar{u}\langle v \rangle$ free names
 $| u(v)$ bound variable
 $| \tau$
 $| \dots$

$P ::= \dots$
 $| (\nu n)$ bound constant
 $| \dots$

disjoint union not present in
the original π -calculus

Name = Const $\uplus$ Var

The notion of free and bound constants and variables are defined as expected:

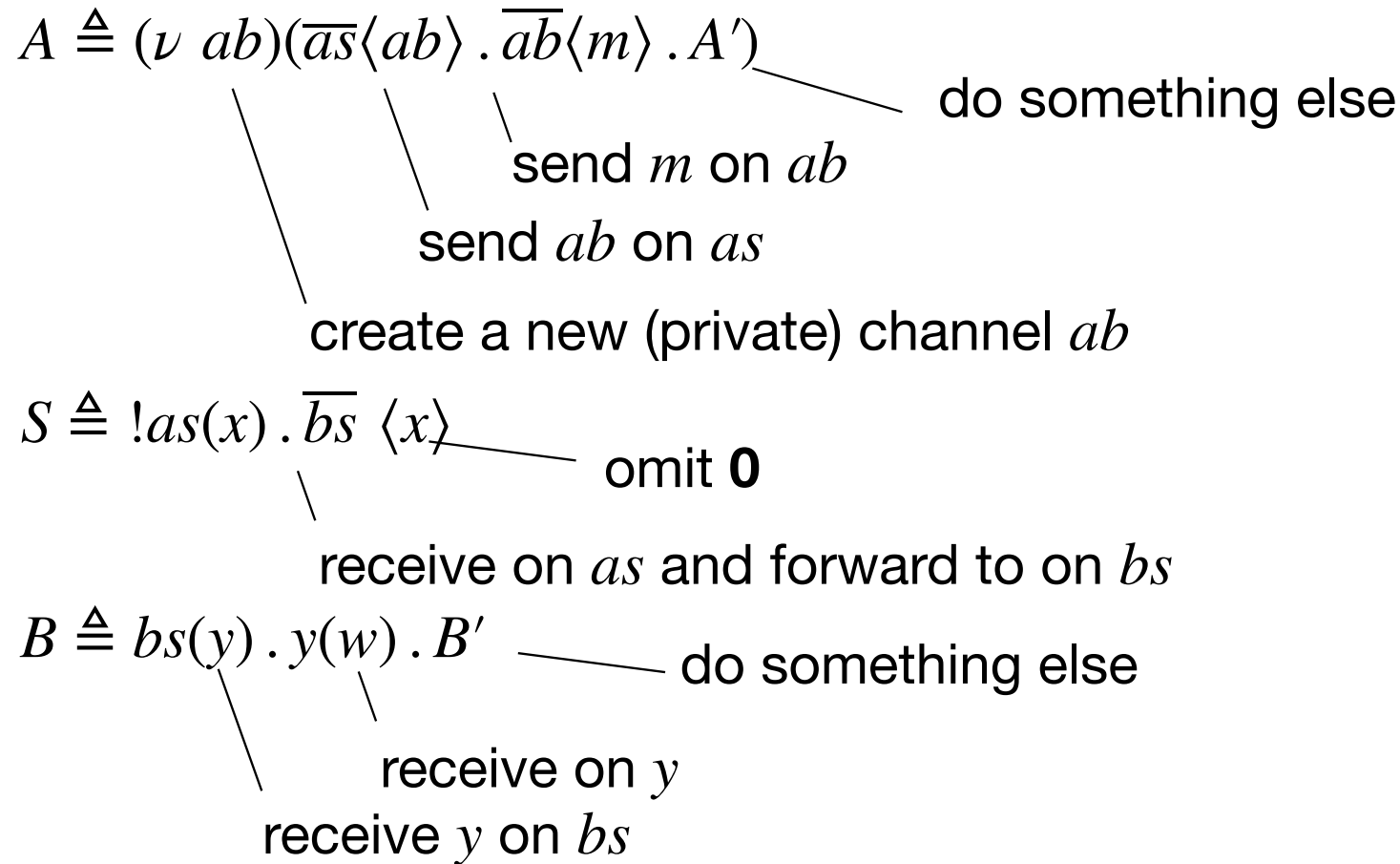
$fc(P)$ and $fv(P)$

π -calculus: syntax details

- The notion of free and bound names (constants and variables) are defined as usual: we focus on $fc(P)$ and $fv(P)$
- The notion of capture avoiding substitution is defined as usual: $P[y/x]$
- Processes are taken up-to alpha-renaming of bound names:
 - $u(y) . P = u(z) . (P[z/y])$ if $z \notin fv(P)$
 - $(\nu n) . P = (\nu n') . (P[n'/n])$ if $n' \notin fc(P)$
- We focus on closed processes, i.e., process P such that $fc(P) = fv(P) = \emptyset$

Examples

$$(\nu \text{ } as)(\nu \text{ } bs)(A \mid S \mid B)$$



Example

the scope
of ab is
extended
to include
both
processes

$$(\nu as)(\nu bs)((\nu ab)(\boxed{\overline{as}\langle ab \rangle} . \overline{ab}\langle m \rangle . A' \mid \boxed{!as(x)} . \overline{bs}\langle x \rangle \mid bs(y) . y(w) . B')$$

scope extrusion

$$(\nu as)(\nu bs)((\nu ab)(\overline{ab}\langle m \rangle . A' \mid \boxed{\overline{bs}\langle ab \rangle}) \mid \boxed{bs(y)} . y(w) . B') \mid S$$

scope extrusion

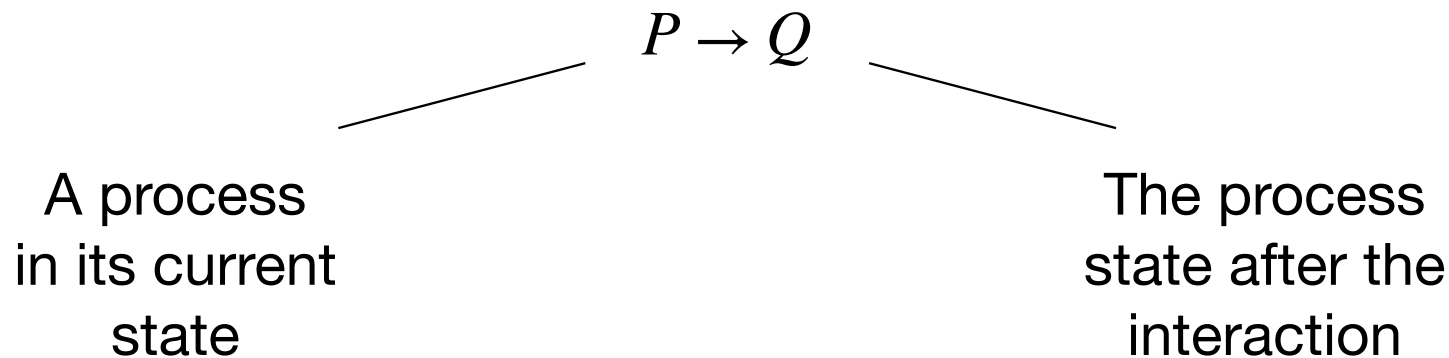
$$(\nu as)(\nu bs)((\nu ab)(\boxed{\overline{ab}\langle m \rangle} . A' \mid \mathbf{0} \mid \boxed{ab(w)} . B') \mid S)$$

$$(\nu as)(\nu bs)((\nu ab)(A' \mid \mathbf{0} \mid B'[m/w]) \mid S)$$

π -calculus: reduction semantics

We define the semantics by:

- a structural congruence relation $\equiv$, and
- a reduction relation $\rightarrow$



Structural congruence

Abelian monoid laws for parallel

$$(P \mid Q) \mid R \equiv P \mid (Q \mid R)$$

$$P \mid Q \equiv Q \mid P$$

$$P \mid \mathbf{0} \equiv \mathbf{0}$$

Scope laws

$$(\nu n)(\nu m)P \equiv (\nu m)(\nu n)P$$

$$(\nu n)\mathbf{0} \equiv \mathbf{0}$$

$$(\nu n)(P \mid Q) \equiv (\nu m)P \mid Q \text{ if } n \notin fc(Q)$$

Matching

$$[x = x]P \equiv P$$

Summands can be permuted in

$$\sum_{i \in I} \pi_i . P_i$$

Unfolding of replication

$$!P \equiv P \mid !P$$

α -renaming

$$P \equiv_{\alpha} Q \Rightarrow P \equiv Q$$

Reduction semantics

$$[Com] (\bar{n}\langle m \rangle . P + P') \mid (n(x) . Q + Q') \rightarrow P \mid Q[m/x]$$

$$[Tau] (\tau . P + Q) \rightarrow P$$

$$[Res] \frac{P \rightarrow P'}{(\nu n)P \rightarrow (\nu n)P'}$$

$$[Par] \frac{P \rightarrow P'}{P \mid Q \rightarrow P' \mid Q}$$

$$[Eq] \frac{P \equiv Q \quad Q \rightarrow Q' \quad Q' \equiv P'}{P \rightarrow P'}$$

0-CFA for π -calculus

CFA predicts communications

